INCH POUND

MIL-M-38510/82C
20 JULY 2005
SUPERSEDING
MIL-M-38510/82B
30 OCTOBER 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, SCHOTTKY TTL, MAGNITUDE COMPARITORS, MONOLITHIC SILICON

Inactive for new design after 8 July 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, magnitude comparator microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.4).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device type. The device type is as follows:

Device type	Circuit
01	4 bit comparator

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat-pack
X	CQCC2-N20	20	Square chip carrier package
2	CQCC1-N20	20	Square chip carrier package

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	605 mW
Lead temperature (soldering 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC})	
Cases E, F, X, and 2	(See MIL-STD-1835)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc 3/
Normalized fanout (each output)	10 maximum
Case operating temperature range (T_C)	-55°C to 125°C

2.0 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short circuit condition (e.g., I_{OS}).
2/ Maximum junction temperature should not be exceeded except in accordance with allowable short duration burn-in screening condition in accordance with MIL-PRF-38535.
3/ 0.7 V at +125 ° C

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.3).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Truth table. The truth table shall be as specified on figure 2.

3.3.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuit shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in Table II. The electrical tests for each subgroup are described in Table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-PRF-38535, appendix A).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ Unless Otherwise Specified	Limits		Unit
			Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V V_{IL} = 0.8 V, I_{OH} = -1 mA 1/	2.5		V
		T_C = +125°C, V_{IL} = 0.7 V			
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.8 V, I_{OL} = 20 mA 1/		0.5	V
		T_C = +125°C, V_{IL} = 0.7 V		0.45	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C		-1.2	V
High level input current (A < B, A > B inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		50	µA
High level input current (all other inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		150	µA
High level input current (all inputs)	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		1	mA
Low level input current (A < B, A > B inputs)	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	-1	-2	mA
Low level input current (all other inputs)	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	-3	-6	mA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 1/ 2/	-40	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 3/		115	mA
Collector cut-off current	I_{CEX}	V_{CC} = 5.5 V, V_{IL} = GND V_{OH} = 5.5 V		250	µA
Propagation delay from any A or B input to A < B or A > B outputs	t_{PLH1}	C_L = 50 pF ± 10%, 3 gate levels R_L = 280 Ω 4/ 5/	2	23	ns
Propagation delay from any A or B input to A = B output	t_{PLH2}	C_L = 50 pF ± 10%, 3 gate levels R_L = 280 Ω 4/ 5/	2	25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ Unless Otherwise Specified	Limits		Unit
			Min	Max	
Propagation delay from A < B or A = B input to A > B output	t_{PLH3}	C_L = 50 pF ± 10%, 2 gate levels R_L = 280 Ω 4/ 5/	2	11.5	ns
Propagation delay from A = B input to A = B output	t_{PLH4}	C_L = 50 pF ± 10%, 1 gate level R_L = 280 Ω 4/ 5/	2	16	ns
Propagation delay from A > B or A = B input to A < B output	t_{PLH5}	C_L = 50 pF ± 10%, 2 gate levels R_L = 280 Ω 4/ 5/	2	11.5	ns
Propagation delay from any A or B input to A > B or A < B output	t_{PHL1}	C_L = 50 pF ± 10%, 3 gate levels R_L = 280 Ω 4/ 5/	2	24	ns
Propagation delay from any A or B input to A = B output	t_{PHL2}	C_L = 50 pF ± 10%, 3 gate levels R_L = 280 Ω 4/ 5/	2	24	ns
Propagation delay from A < B or A = B inputs to A > B output	t_{PHL3}	C_L = 50 pF ± 10%, 2 gate levels R_L = 280 Ω 4/ 5/	2	13	ns
Propagation delay from A = B input to A = B output	t_{PHL4}	C_L = 50 pF ± 10%, 1 gate level R_L = 280 Ω 4/ 5/	2	12	ns
Propagation delay from A > B or A = B input to A < B output	t_{PHL5}	C_L = 50 pF ± 10%, 2 gate levels R_L = 280 Ω 4/ 5/	2	13	ns

1/ All unspecified inputs grounded.

2/ Not more than one output should be shorted at a time.

3/ I_{CC} is measured with outputs open, A = B grounded, and all other inputs at 4.5 V.

4/ The longest gate level propagation delay paths are not tested (5 gate levels); however, they can be calculated: $t = t_{P1}$ maximum + t_{P2} maximum – t_{P4} minimum.

5/ Gate level references are made with respect to the circuit logic diagram.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 Test requirements	Subgroups (see table III)	
	Class S Devices	Class B Devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 8, 9, 10, 11
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3, 7, 8, 9, 10, 11	N/A
Group C end-point electrical Parameters	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3
Group D end point electrical Parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.3 Screening. Screening shall be in accordance with MIL-PRF-38535, and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

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4.4 Technology Conformance Inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

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Terminal number		Terminal name	
Cases		Device type 01	
2, X	E, F	2, X	E, F
1	1	NC	B3
2	2	B3	A < B In
3	3	A < B In	A = B In
4	4	A = B In	A > B In
5	5	A > B In	A > B Out
6	6	NC	A = B Out
7	7	A > B Out	A < B Out
8	8	A = B Out	GND
9	9	A < B Out	B0
10	10	GND	A0
11	11	NC	B1
12	12	B0	A1
13	13	A0	A2
14	14	B1	B2
15	15	A1	A3
16	16	NC	V_{CC}
17		A2	
18		B2	
19		A3	
20		V_{CC}	

FIGURE 1. Terminal connections.

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Device Type 01

Comparing Inputs				Cascading inputs			Outputs		
A3, B3	A2, B2	A1, B1	A0, B0	A > B	A < B	A = B	A > B	A < B	A = B
A3 > B3	X	X	X	X	X	X	H	L	L
A3 < B3	X	X	X	X	X	X	L	H	L
A3 = B3	A2 > B2	X	X	X	X	X	H	L	L
A3 = B3	A2 < B2	X	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 > B1	X	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 < B1	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 > B0	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 < B0	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	L	L	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	H	L	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	H	L	L	H
A3 = B3	A2 = B2	A1 = B1	A0 = B0	X	X	H	L	L	H
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	H	L	L	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	L	H	H	L

H = High Level

L = Low Level

X = Irrelevant

FIGURE 2. Truth Table.

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DEVICE TYPE 01

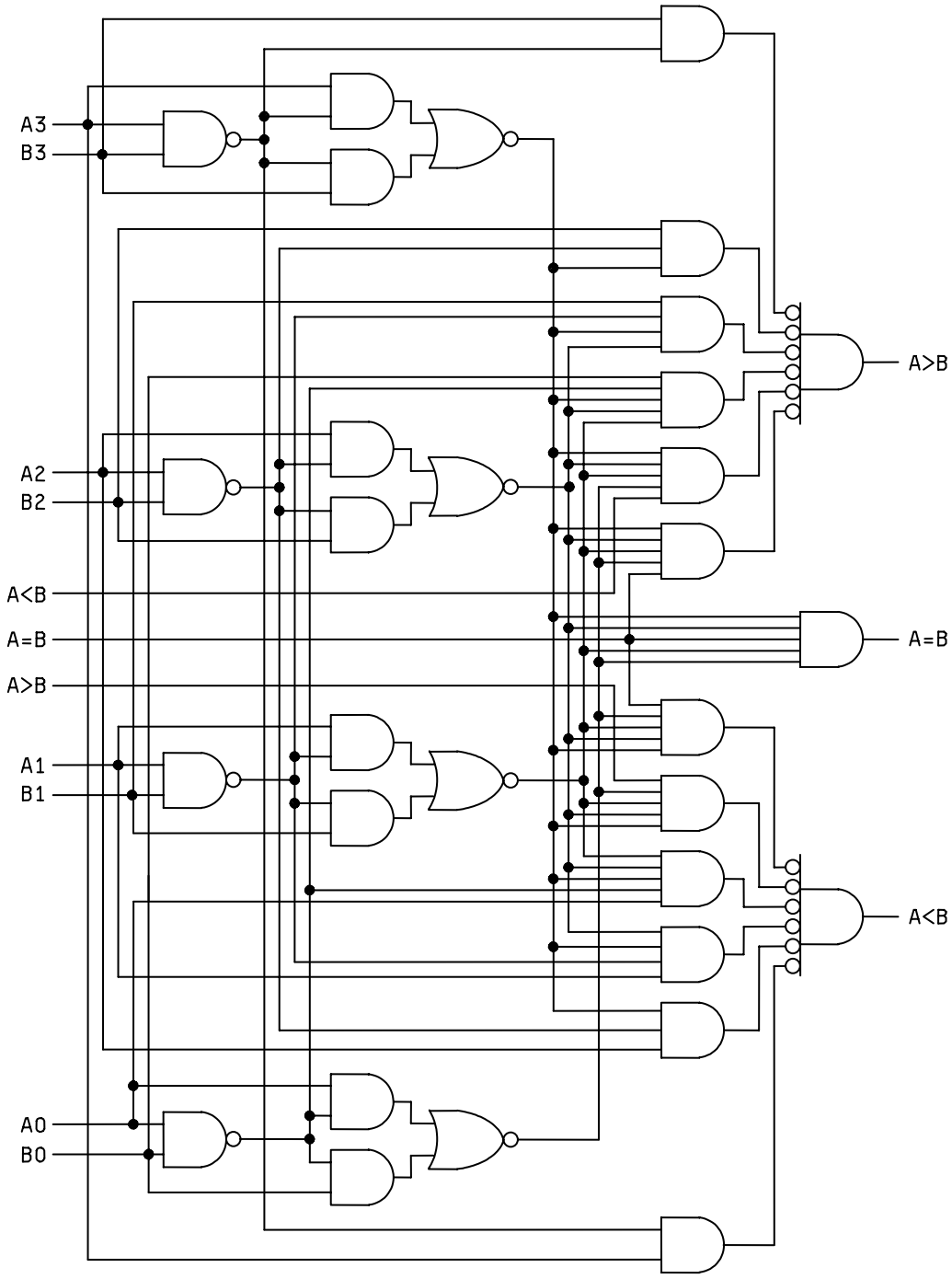
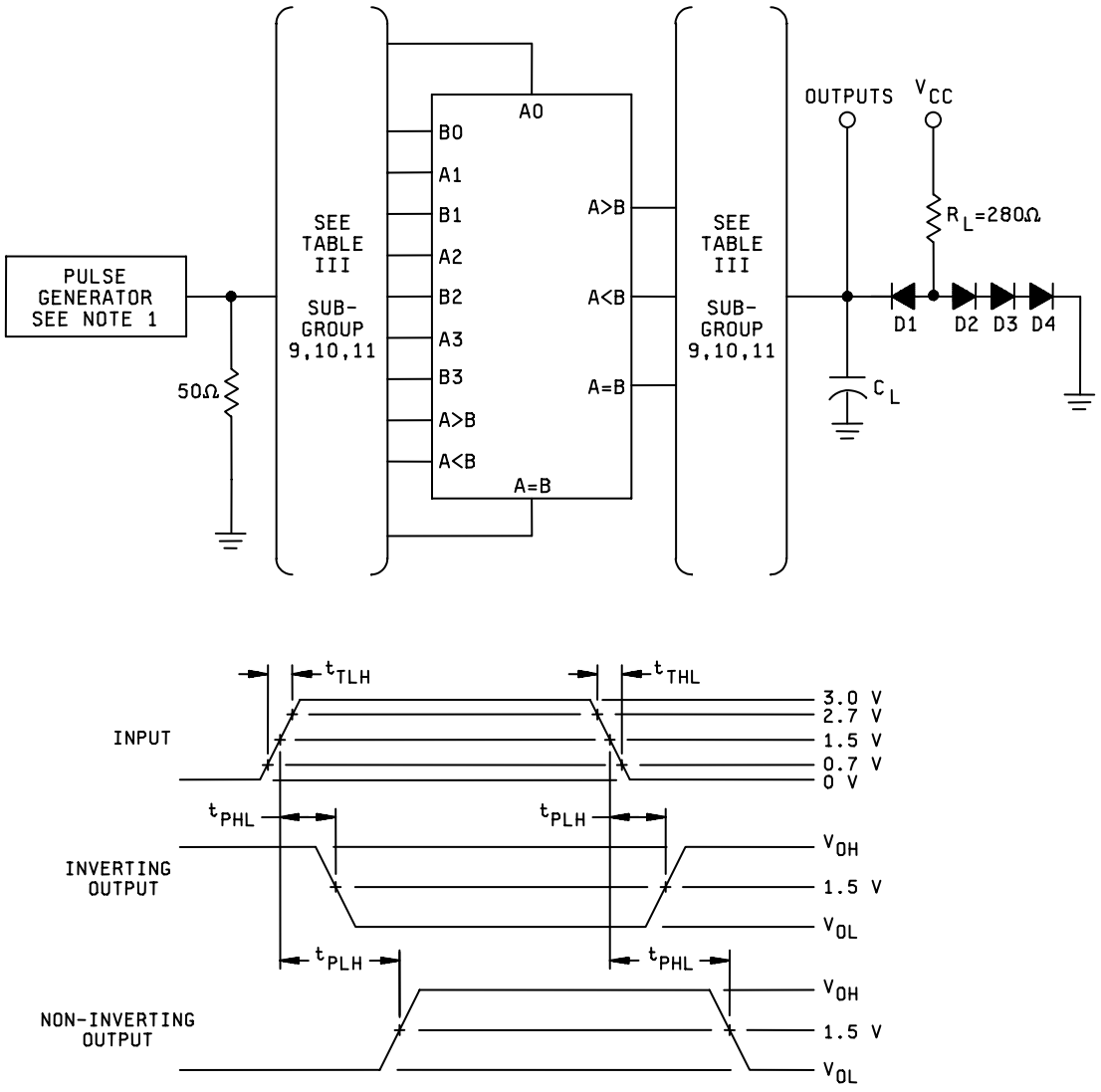


FIGURE 3. Logic diagram.

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DEVICE TYPE 01



NOTES:

1. The pulse generator has the following characteristics: V_{gen} = 3 V, t_{TLH} and t_{THL} ≤ 2.5 ns, t_p = 500 ns, PRR ≤ 1 MHz, and Z_{out} ≈ 50 Ω.
2. CL = 50 PF ± 10 percent and includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.

FIGURE 4. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
1 T_C=+25°C	V_{OH}	3006	1 2 3	0.8 V 0.8 V GND	0.8 V	0.8 V 2.0 V 0.8 V	0.8 V	-1 mA	-1 mA	-1 mA	GND “ “	0.8 V 0.8 V GND	GND 0.8 V 0.8 V	0.8 V 0.8 V GND	GND 0.8 V 0.8 V	GND 0.8 V 0.8 V	0.8 V 0.8 V GND	GND 0.8 V 0.8 V	4.5 V “ “	A>B OUT A=B OUT A<B OUT	2.5 “ “		V dc “ “
	V_{OL}	3007	4 5 6	2.0 V 2.0 V 0.8 V	2.0 V	GND 0.8 V 2.0 V	GND	20 mA	20 mA	20 mA	“ “ “	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	2.0 V 2.0 V 0.8 V	“ “ “	A>B OUT A=B OUT A<B OUT		0.5 “ “	“ “ “
	V_{IC}		7 8 9 10 11 12 13 14 15 16 17	-18 mA	-18 mA	-18 mA	-18 mA				“ “ “ “ “ “ “ “ “ “ “	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	“ “ “ “ “ “ “ “ “ “ “	B3 A<B IN A=B IN A>B IN B0 A0 B1 A1 A2 B2 A3		-1.2 “ “ “ “ “ “ “ “ “ “	“ “ “ “ “ “ “ “ “ “ “
	I_{IH1}	3010	18 19	GND GND	2.7 V		2.7 V				“ “	GND GND	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V “	A<B IN A>B IN		50 50	µA “
	I_{IH2}		20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36	2.7 V GND “ “ “ “ “ “ “ 2.7 V GND “ “ “ “ “ 5.5 V		2.7 V					“ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “	GND 2.7 V GND “ “ “ “ “ 2.7 V 5.5 V GND “ “ “ “ “	5.5 V GND 2.7 V 5.5 V “ “ “ “ 5.5 V GND “ “ “ “ “ “	5.5 V “ “ 2.7 V “ “ “ “ GND GND 2.7 V 5.5 V GND “ “ “	5.5 V “ “ “ 2.7 V GND 5.5 V “ “ “ “ “ “ “ “ “	5.5 V “ “ “ “ 2.7 V GND “ “ “ “ “ “ “ “ “	5.5 V GND 2.7 V “ “ “ “ 2.7 V GND “ “ “ “ “ “ “	GND 5.5 V “ “ “ “ “ “ 2.7 V 5.5 V “ “ “ “ “ “ 2.7 V	“ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “	B3 A=B IN B0 A0 B1 A1 A2 B2 A3 B3 B0 A0 B1 A1 A2 B2 A3		150 “ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “	“ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “ “

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
1 T_C=+25°C	I_{IH3}	3010	37	GND	5.5 V			5.5 V			GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A<B IN		1	mA
			38	GND							"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A>B IN		"	"
			39	5.5 V							"							GND	"	B3		"	"
			40	GND		5.5 V					"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A=B IN		"	"
			41	"							"	5.5 V	GND	"	"	"	GND	"	"	B0		"	"
			42	"							"	GND	5.5 V	"	"	"	"	"	"	A0		"	"
			43	"							"	"	"	"	"	"	"	"	"	B1		"	"
			44	"							"	"	"	GND	"	"	"	"	"	A1		"	"
			45	"							"	"	"	"	GND	"	"	"	"	A2		"	"
			46	"							"	"	"	"	5.5 V	GND	5.5 V	"	"	B2		"	"
			47	"							"							"	"	A3		"	"
			48	5.5 V							"							"	"	B3		"	"
			49	GND							"	5.5 V	5.5 V	GND	5.5 V	5.5 V	GND	"	"	B0		"	"
			50	"							"	5.5 V	"	GND	5.5 V	"	"	"	"	A0		"	"
			51	"							"	GND	"	5.5 V	GND	"	"	"	"	B1		"	"
			52	"							"	"	"	5.5 V	5.5 V	"	"	"	"	A1		"	"
			53	"							"	"	"	GND	"	"	5.5 V	"	"	A2		"	"
			54	"							"	"	"	GND	"	"	5.5 V	"	"	B2		"	"
			55	5.5 V							"							"	"	A3		"	"
	I_{IL1}	3009	56	5.5 V	0.5 V						"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A<B IN	-1.0	-2.0	mA
			57	5.5 V			0.5 V				"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A>B IN	-1.0	-2.0	"
	I_{IL2}		58	0.5 V							"							"	"	B3	-3.0	-6.0	"
			59	5.5 V		0.5 V					"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A=B IN	"	"	"
			60	"							"	0.5 V	5.5 V	"	"	"	"	"	"	B0	"	"	"
			61	"							"	5.5 V	0.5 V	"	"	"	"	"	"	A0	"	"	"
			62	"							"	"	5.5 V	0.5 V	"	"	"	"	"	B1	"	"	"
			63	"							"	"	"	5.5 V	"	"	"	"	"	A1	"	"	"
			64	"							"	"	"	"	0.5 V	"	"	"	"	A2	"	"	"
			65	"							"	"	"	"	5.5 V	0.5 V	"	"	"	B2	"	"	"
			66	"							"	"	"	"	5.5 V	5.5 V	0.5 V	"	"	A3	"	"	"
	I_{OS}	3011	67	GND	GND	GND		GND			"	GND	GND	GND	GND	GND	GND	GND	"	A>B OUT	-40	-100	"
			68	"		2.0 V			GND		"	"	"	"	"	"	"	"	"	A=B OUT	"	"	"
			69	"		GND	GND			GND	"	"	"	"	"	"	"	"	"	A<B OUT	"	"	"
	I_{CC}	3005	70	4.5 V	4.5 V	GND	4.5 V				"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	V_{CC}		115	"
	I_{CEX}		71	GND	GND	GND		5.5 V			"	GND	GND	GND	GND	GND	GND	GND	"	A>B OUT		250	µA
			72	"		5.5 V			5.5 V		"	"	"	"	"	"	"	"	"	A=B OUT		"	"
			73	"		GND	GND			5.5 V	"	"	"	"	"	"	"	"	"	A<B OUT		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted. V_{IL} = 0.7 V and V_{OL} (max) = 0.45 V.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C=+25°C	Func-tional Test 4/	3014 2/	74	B	A	B	B	H 3/	L 3/	L 3/	GND	A	B	A	B	B	A	A	4.5 V	2/			
			75	A	B	"	A	L	"	H	"	B	A	B	A	A	B	B	"				
			76	"	A	"	B	H	"	L	"	A	B	A	B	A	B	A	"				
			77	"	B	"	A	L	"	H	"	B	A	B	A	B	A	"	"				
			78	"	A	"	B	H	"	L	"	A	B	B	A	A	"	"	"				
			79	"	B	"	A	L	"	H	"	B	A	A	B	"	"	"	"				
			80	"	A	"	B	H	"	L	"	B	A	"	A	"	"	"	"				
			81	"	B	"	A	L	"	H	"	A	B	"	"	"	"	"	"				
			82	"	B	"	A	H	"	L	"	"	A	"	"	"	"	"	"				
			83	"	"	A	B	L	H	"	"	"	"	"	"	"	"	"	"				
			84	"	"	A	A	"	H	"	"	"	"	"	"	"	"	"	"				
			85	"	A	B	B	"	L	H	"	"	"	"	"	"	"	"	"				
			86	"	"	B	A	"	L	L	"	"	"	"	"	"	"	"	"				
			87	"	"	A	B	"	H	"	"	"	"	"	"	"	"	"	"				
			88	"	"	A	A	"	H	"	"	"	"	"	"	"	"	"	"				
			89	"	B	B	B	H	L	H	"	"	"	'	"	"	"	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C 4/.																						
9 T_C=+25°C	t_{PLH1}	3003 (Fig. 4)	90	IN		GND	GND			OUT	GND	GND	GND	GND	GND	GND	GND	2.7 V	5.0 V	B3 to A<B	2	19	ns
			91	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A<B	"	"	"
			92	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A>B	"	"	"
			93	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A<B	"	"	"
			94	"	GND	"		OUT			"	"	"	2.7 V	IN	GND	"	"	"	A1 to A>B	"	"	"
			95	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A>B	"	"	"
			96	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A<B	"	"	"
			97	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A>B	"	"	"
	t_{PLH2}		98	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B3 to A=B	"	21	"
			99	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A=B	"	"	"
			100	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A=B	"	"	"
			101	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A=B	"	"	"
			102	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A=B	"	"	"
			103	"		"			"		"	"	"	"	GND	IN	"	"	"	A2 to A=B	"	"	"
			104	"		"			"		"	"	"	"	"	GND	IN	"	"	B2 to A=B	"	"	"
			105	"		"			"		"	"	"	"	"	GND	GND	IN	"	A3 to A=B	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
9 T_C=+25°C	t_{PLH3}	3003 (Fig. 4)	106	GND	IN	GND		OUT			GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	A<B to A>B	2	9.5	ns
			107	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A=B to A>B	"	"	"
	t_{PLH5}		108	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A=B to A<B	"	"	"
			109	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A>B to A<B	"	"	"
	t_{PLH4}		110	"		IN			OUT		"	"	"	"	"	"	"	"	"	A=B to A=B	"	13	"
	t_{PHL1}		111	IN		GND	GND			OUT	"	"	"	"	"	"	"	2.7 V	"	B3 to A<B	"	19.5	"
			112	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A<B	"	"	"
			113	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A>B	"	"	"
			114	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A<B	"	"	"
			115	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A1 to A>B	"	"	"
			116	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A>B	"	"	"
			117	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A<B	"	"	"
			118	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A>B	"	"	"
	t_{PHL2}		119	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B3 to A=B	"	"	"
			120	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A=B	"	"	"
			121	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A=B	"	"	"
			122	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A=B	"	"	"
			123	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A=B	"	"	"
			124	"		"			"		"	"	"	"	GND	IN	"	"	"	A2 to A=B	"	"	"
			125	"		"			"		"	"	"	"	"	GND	IN	"	"	B2 to A=B	"	"	"
			126	"		"			"		"	"	"	"	"	"	GND	IN	"	A3 to A=B	"	"	"
	t_{PHL3}		127	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A<B to A>B	"	10.5	"
			128	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A=B to A>B	"	"	"
	t_{PHL5}		129	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A=B to A<B	"	"	"
			130	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A>B to A<B	"	"	"
	t_{PHL4}		131	"		IN			OUT		"	"	"	"	"	"	"	"	"	A=B to A=B	"	9.5	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
10 T_C=+125°C	t_{PLH1}	3003 (Fig. 4)	132	IN		GND	GND			OUT	GND	GND	GND	GND	GND	GND	GND	2.7 V	5.0 V	B3 to A<B	2	23	ns
			133	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A<B	"	"	"
			134	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A>B	"	"	"
			135	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A<B	"	"	"
			136	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A1 to A>B	"	"	"
			137	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A>B	"	"	"
			138	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A<B	"	"	"
			139	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A>B	"	"	"
	t_{PLH2}		140	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B3 to A=B	"	25	"
			141	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A=B	"	"	"
			142	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A=B	"	"	"
			143	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A=B	"	"	"
			144	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A=B	"	"	"
			145	"		"					"	"	"	"	GND	IN	"	"	"	A2 to A=B	"	"	"
			146	"		"					"	"	"	"	"	GND	IN	"	"	B2 to A=B	"	"	"
			147	"		"					"	"	"	"	"	"	GND	IN	"	A3 to A=B	"	"	"
	t_{PLH3}		148	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A<B to A>B	"	11.5	"
			149	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A=B to A>B	"	"	"
	t_{PLH5}		150	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A=B to A<B	"	"	"
			151	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A>B to A<B	"	"	"
	t_{PLH4}		152	"		IN			OUT		"	"	"	"	"	"	"	"	"	A=B to A=B	"	16	"
	t_{PHL1}		153	IN		GND	GND			OUT	"	"	"	"	"	"	"	2.7 V	"	B3 to A<B	"	24	"
			154	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A<B	"	"	"
			155	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A>B	"	"	"
			156	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A<B	"	"	"
			157	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A1 to A>B	"	"	"
			158	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A>B	"	"	"
			159	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A<B	"	"	"
			160	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A>B	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B3	A<B IN	A=B IN	A>B IN	A>B OUT	A=B OUT	A<B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}		Min	Max	
10 T_C=+125°C	t_{PHL2}	3003 (Fig. 4)	161	IN		2.7 V			OUT		GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	B3 to A=B	2	24	ns
			162	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A=B	"	"	"
			163	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A=B	"	"	"
			164	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A=B	"	"	"
			165	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A=B	"	"	"
			166	"		"			"		"	"	"	"	GND	IN	"	"	"	A2 to A=B	"	"	"
			167	"		"			"		"	"	"	"	"	GND	IN	"	"	B2 to A=B	"	"	"
			168	"		"			"		"	"	"	"	"	"	GND	IN	"	A3 to A=B	"	"	"
	t_{PHL3}		169	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A<B to A>B	"	13	"
			170	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A=B to A>B	"	"	"
	t_{PHL5}		171	"		GND	GND			OUT	"	"	"	"	"	"	"	"	"	A=B to A<B	"	"	"
			172	"		IN	IN			OUT	"	"	"	"	"	"	"	"	"	A>B to A<B	"	"	"
	t_{PHL4}		173	"		IN			OUT		"	"	"	"	"	"	"	"	"	A=B to A=B	"	12	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ Cases X and 2 pins not designated are NC.

2/ Tests shall be performed in sequence.

3/ Output voltages shall be either (a) H = 2.5 V, minimum and L = 0.5 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V when using a high speed checker single comparator.

4/ A = 2.5 V and B = 0.5 V for T_C = +25°C and -55°C. A = 2.5 V and B = 0.45 V for T_C = +125°C.

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory)

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. PIN and compliance identifier, if applicable (see 1.2).
c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
d. Requirement for certificate of compliance, if applicable.
e. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.
f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.
i. Requirements for "JAN" marking.
j. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

6.4 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

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6.5 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535 and MIL-HDBK-1331, and as follows:

GND ..... Electrical ground (common terminal)
I_{IN} ..... Current flowing into an input terminal.
V_{IN} ..... Voltage level at an input terminal.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming shall not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic industry type. Generic industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Device type	Commercial type
01	4S85

6.8 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table I.

Table IV. Manufacturers' designations

Device Type	Texas Instruments	Signetics Corporation
	Circuits	
	A	B
01	X	X

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army – CR
Navy - EC
Air Force – 11
DLA-CC

Preparing activity:
DLA - CC

(Project 5962-2129)

Review activities:
Army – MI, SM
Navy – AS, CG, MC, SH, TD
Air Force – 03, 19, 99

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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